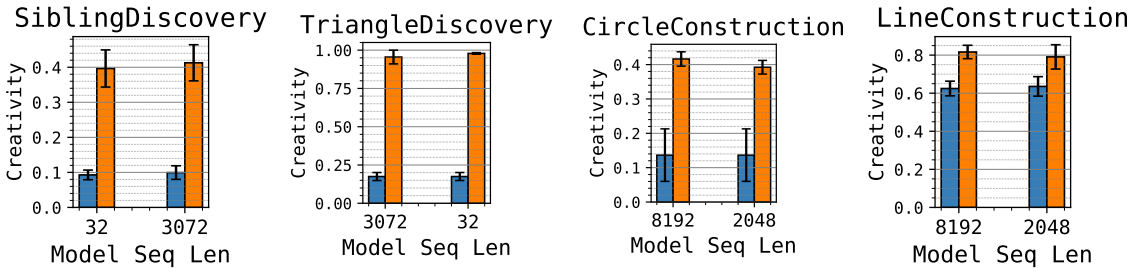

Figure 17. **Batch size and algorithmic creativity for `Gemma v1 (2B)`:** Algorithmic creativity increases under multi-token prediction across various batch sizes. Note that here batch size is effectively proportional to the model sequence length, since we pack multiple finetuning examples into the sequence.

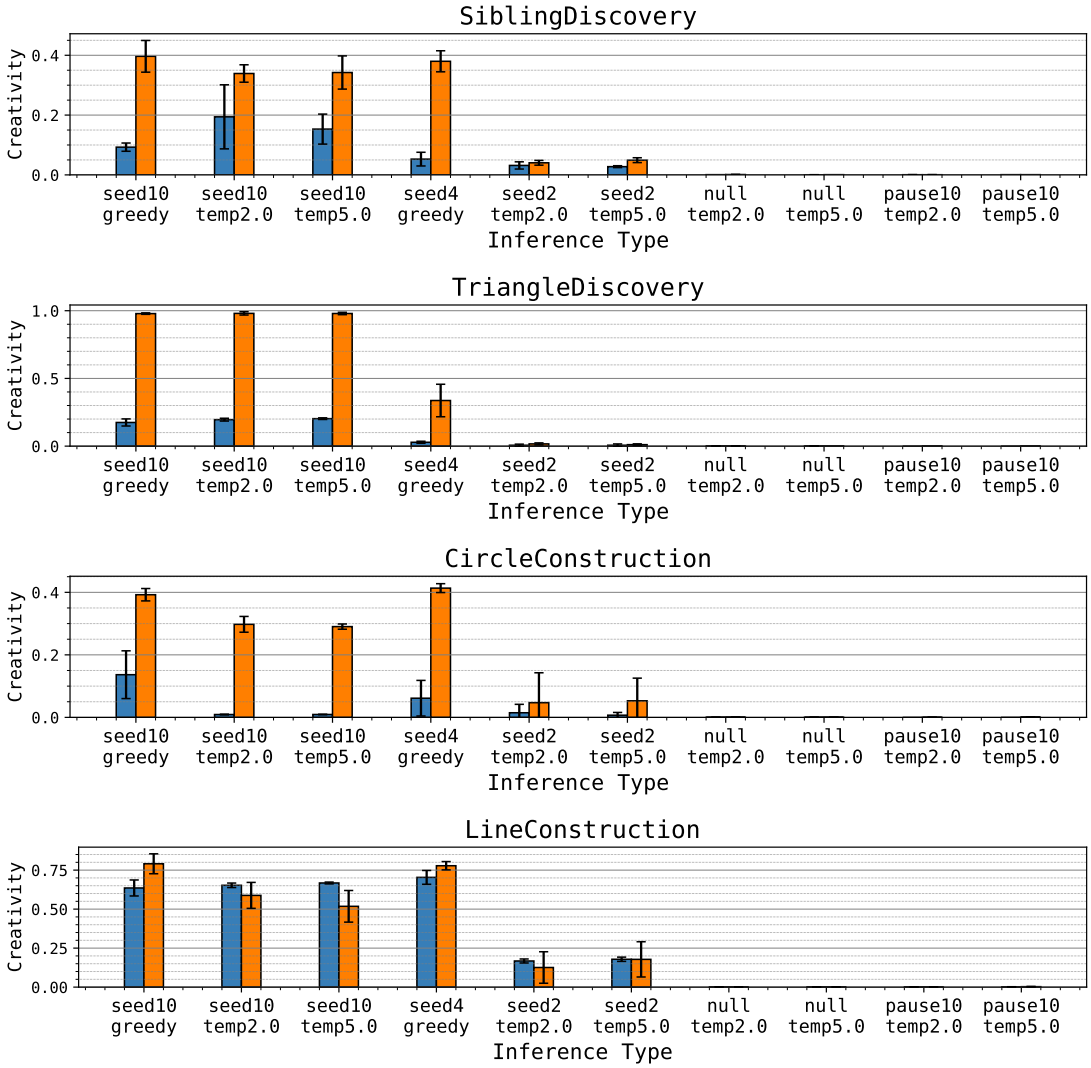

Figure 18. **Algorithmic creativity under various sampling conditions for `Gemma v1 (2B)`:** Across all conditions, and in almost all datasets (with a few exceptions in `Line Construction`), multi-token prediction improves creativity. Furthermore, seed-conditioning achieves best algorithmic creativity, with a longer seed helping more.

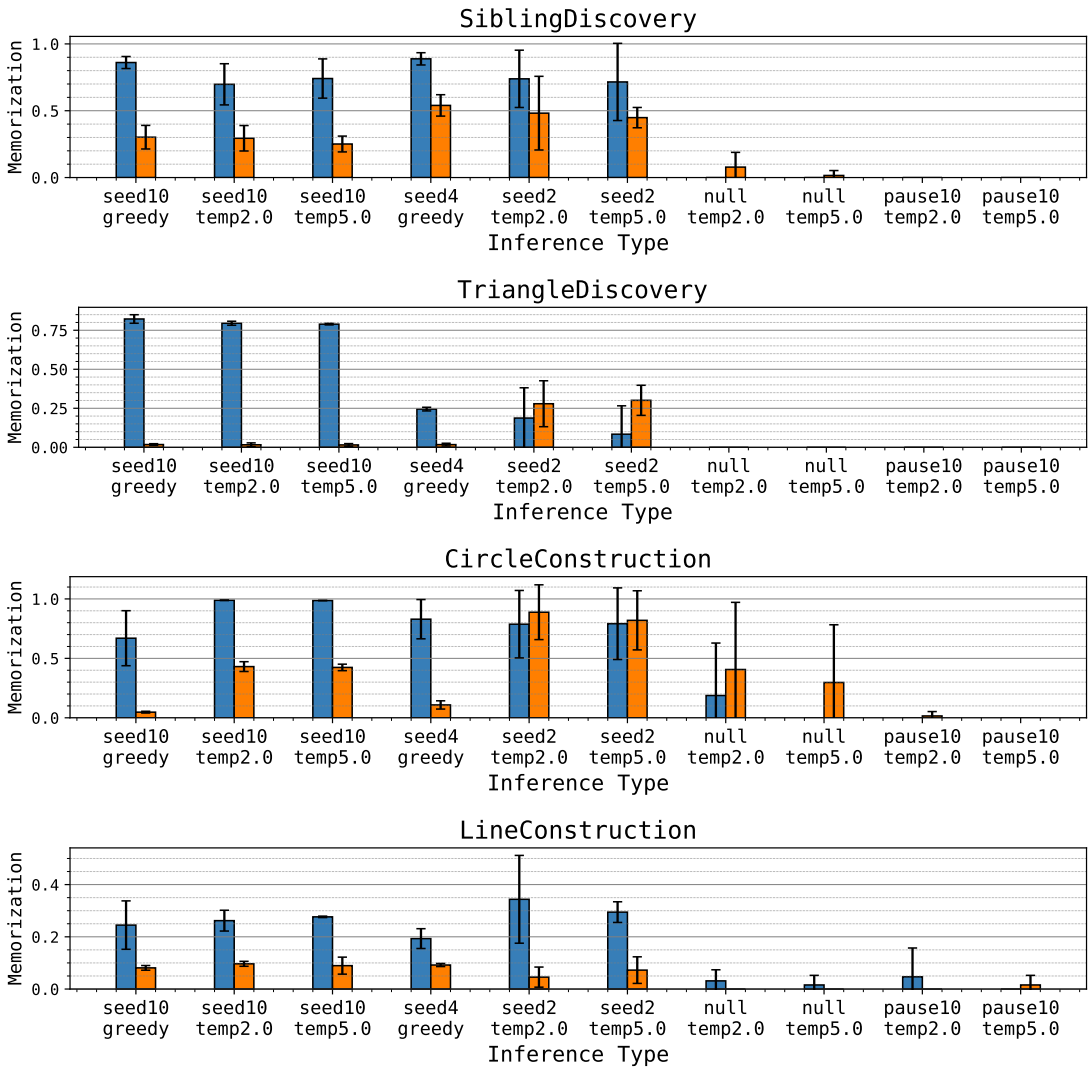

Figure 19. **Memorization under various sampling conditions for `Gemma v1 (2B)`:** Barring a few conditions, the most prominent trend is that memorization reduces under multi-token prediction for various sampling conditions. Observe that the null and pause-conditioned models *do* produce some memorized output while their creativity was non-existent.

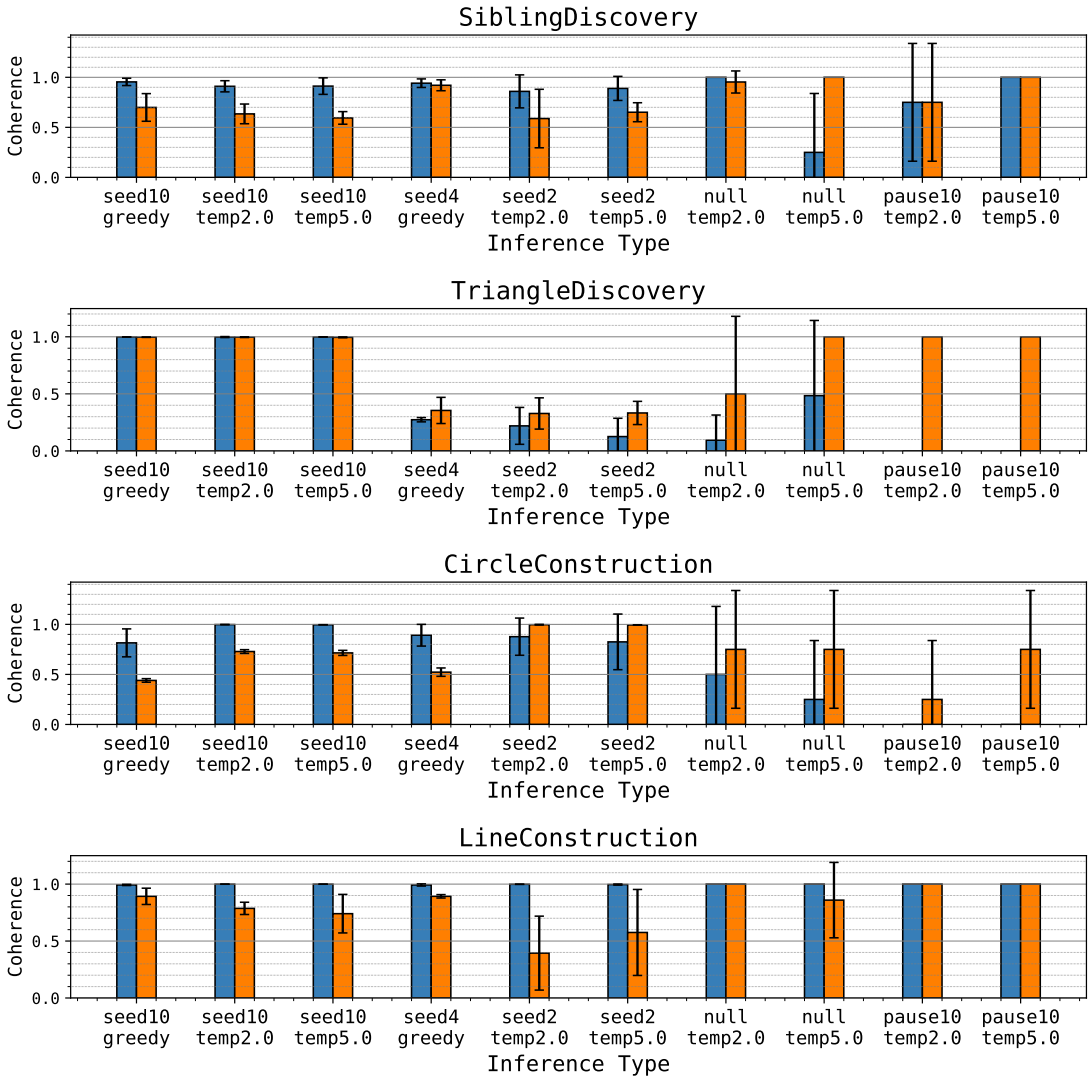

Figure 20. **Coherence under various sampling conditions for `Gemma v1 (2B)`:** Surprisingly, coherence of all models is high or at least noticeable, across various sampling conditions. This suggests that the low algorithmic creativity of the null-conditioned models in the previous plots arises from model collapsing to single original point.